Assignment 10 — Problem Solutions

Two-Sample Inferences and Goodness-of-Fit Tests

Name: _____

Chapter 9: Two-Sample Inferences

Problem 9.2.6 (Authors and alcohol: pooled t test, one-sided)

Given. Let X = age at death of authors noted for alcohol abuse ($n = 9$) and Y = age at death of authors not noted ($m = 12$). The pooled standard deviation is given as $s_p = 13.9$.

Step 1. Hypotheses and significance level

$$H_0 : \ \mu_X = \mu_Y \qquad \text{vs.} \qquad H_1 : \ \mu_X < \mu_Y, \qquad \alpha = 0.05.$$

The one-sided alternative reflects the claim that problem drinkers die younger.

Step 2. Test statistic (Theorem 9.2.2)

Assuming equal variances, under H_0

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{\dfrac{1}{n} + \dfrac{1}{m}}} \ \sim \ t_{\,n+m-2}.$$

Step 3. Sample means

$$\sum x_i = 48 + 66 + 71 + 65 + 56 + 67 + 67 + 70 + 77 = 587, \qquad \bar{x} = \frac{587}{9} = 65.222,$$

$$\sum y_i = 65 + 87 + 32 + 77 + 89 + 86 + 77 + 84 + 64 + 88 + 90 + 67 = 906, \quad \bar{y} = \frac{906}{12} = 75.500.$$

Step 4. Standard error

$$\sqrt{\frac{1}{9} + \frac{1}{12}} = \sqrt{0.11111 + 0.08333} = \sqrt{0.19444} = 0.44096, \qquad SE = 13.9\,(0.44096) = 6.129.$$

Step 5. Observed test statistic

$$t = \frac{65.222 - 75.500}{6.129} = \frac{-10.278}{6.129} = -1.68.$$

Step 6. Critical region

Degrees of freedom: $n + m - 2 = 9 + 12 - 2 = 19$. For a left-tailed test, reject H_0 if $t \le -t_{.05,19} = -1.729$.

Step 7. Decision

Since $-1.68 > -1.729$, the statistic is not in the rejection region. The P-value is $P(T_{19} \le -1.68) \approx 0.055 > 0.05$.

Conclusion. Fail to reject H_0. At $\alpha = 0.05$, a difference of about ten years in mean lifespan is not quite statistically significant (it is borderline, $p \approx 0.055$). The large spread in the sober group (Bourne died at 32) inflates the standard error.

Problem 9.2.12 (Z test with known variances: find the P-value)

Given. $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$; $\sigma_X^2 = 17.6$, $\sigma_Y^2 = 22.9$ (known); $n = 10$, $m = 20$; $\bar{x} = 81.6$, $\bar{y} = 79.9$.

Step 1. Test statistic

With known variances, under H_0,

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\dfrac{\sigma_X^2}{n} + \dfrac{\sigma_Y^2}{m}}} \sim N(0,1).$$

Step 2. Variance of the difference

$$\frac{\sigma_X^2}{n} = \frac{17.6}{10} = 1.760, \qquad \frac{\sigma_Y^2}{m} = \frac{22.9}{20} = 1.145, \qquad \sqrt{1.760 + 1.145} = \sqrt{2.905} = 1.7044.$$

Step 3. Observed Z

$$z = \frac{81.6 - 79.9}{1.7044} = \frac{1.7}{1.7044} = 0.997.$$

Step 4. P-value (two-sided alternative)

$$P = 2\,P(Z \ge |z|) = 2\,[1 - \Phi(0.997)] = 2\,(1 - 0.8406) = 2(0.1594) = 0.319.$$

Conclusion. The P-value is $\approx \mathbf{0.319}$. This is far above any usual α, so the data give no evidence that $\mu_X \neq \mu_Y$.

Problem 9.2.20 (Mortgages: Welch t test, variances not assumed equal)

Given. X = 30-year fixed rates, Y = ARM rates; $n = m = 10$; $\alpha = 0.01$.

Step 1. Hypotheses

$$H_0:\ \mu_X = \mu_Y \qquad \text{vs.} \qquad H_1:\ \mu_X > \mu_Y, \qquad \alpha = 0.01.$$

ARM rates are expected to be *lower*, i.e. fixed rates higher.

Step 2. Test statistic (Theorem 9.2.3)

Without assuming $\sigma_X^2 = \sigma_Y^2$,

$$T' = \frac{\bar{X} - \bar{Y}}{\sqrt{\dfrac{S_X^2}{n} + \dfrac{S_Y^2}{m}}} \overset{\text{approx}}{\sim} t_\nu, \qquad \nu = \frac{\left(\dfrac{s_X^2}{n} + \dfrac{s_Y^2}{m}\right)^2}{\dfrac{(s_X^2/n)^2}{n-1} + \dfrac{(s_Y^2/m)^2}{m-1}}.$$

Step 3. Means

$$\sum x_i = 36.725 \Rightarrow \bar{x} = 3.6725, \qquad \sum y_i = 33.382 \Rightarrow \bar{y} = 3.3382, \qquad \bar{x} - \bar{y} = 0.3343.$$

Step 4. Variances

$$\sum x_i^2 = 135.0845, \quad SS_X = 135.0845 - \frac{(36.725)^2}{10} = 0.21193, \quad s_X^2 = \frac{0.21193}{9} = 0.023548,$$
$$\sum y_i^2 = 111.8545, \quad SS_Y = 111.8545 - \frac{(33.382)^2}{10} = 0.41867, \quad s_Y^2 = \frac{0.41867}{9} = 0.046519.$$

Step 5. Standard error and observed t

$$\frac{s_X^2}{n} = 0.0023548, \quad \frac{s_Y^2}{m} = 0.0046519, \quad SE = \sqrt{0.0070067} = 0.08371, \quad t' = \frac{0.3343}{0.08371} = 3.99.$$

Step 6. Degrees of freedom

$$\nu = \frac{(0.0070067)^2}{\dfrac{(0.0023548)^2}{9} + \dfrac{(0.0046519)^2}{9}} = \frac{4.909 \times 10^{-5}}{6.16 \times 10^{-7} + 2.404 \times 10^{-6}} = \frac{4.909 \times 10^{-5}}{3.020 \times 10^{-6}} = 16.25 \ \approx \ 16.$$

Step 7. Critical region and decision

Reject H_0 if $t' \geq t_{.01,16} = 2.583$. Since $3.99 > 2.583$ (the P-value is about 0.0005), we reject.

Conclusion. Reject H_0 at $\alpha = 0.01$. Fixed-rate mortgages have significantly higher average rates than ARMs.

Problem 9.3.4 (Mice and magnetic fields: F test for equal variances)

Given. X = weight gain, in magnetic field ($n = 10$, $s_X = 5.67$); Y = control ($m = 10$, $s_Y = 3.18$); $\alpha = 0.05$.

Step 1. Hypotheses

$$H_0:\ \sigma_X^2 = \sigma_Y^2 \qquad \text{vs.} \qquad H_1:\ \sigma_X^2 \neq \sigma_Y^2, \qquad \alpha = 0.05.$$

Step 2. Test statistic (Theorem 9.3.1)

Under H_0,

$$F = \frac{S_Y^2}{S_X^2} \ \sim\ F_{m-1,\,n-1} = F_{9,9}.$$

Reject H_0 if $F \le F_{.025,9,9}$ or $F \ge F_{.975,9,9}$.

Step 3. Sample variances (from the raw data)

$$\sum x_i = 172.3,\ \sum x_i^2 = 3258.13,\quad SS_X = 3258.13 - \frac{(172.3)^2}{10} = 289.401,\quad s_X^2 = \frac{289.401}{9} = 32.156\ (s_X = 5.67),$$

$$\sum y_i = 250.0,\ \sum y_i^2 = 6340.76,\quad SS_Y = 6340.76 - \frac{(250)^2}{10} = 90.76,\qquad s_Y^2 = \frac{90.76}{9} = 10.084\ (s_Y = 3.18).$$

Step 4. Observed F

$$F = \frac{s_Y^2}{s_X^2} = \frac{10.084}{32.156} = 0.314.$$

Step 5. Critical values

$$F_{.025,9,9} = 0.248, \qquad F_{.975,9,9} = 4.026.$$

Step 6. Decision

$0.248 < 0.314 < 4.026$, so F is not in the rejection region (two-sided $P \approx 0.099$).

Conclusion. Fail to reject H_0. There is no statistically significant difference in the variances of weight gain for mice in the magnetic field and controls at $\alpha = 0.05$.

Problem 9.4.4 (Flying saucers: two-proportion Z test)

Given. Let p_S = true probability a sighting is "saucer on ground" in Spain, p_{NS} = same outside Spain.

	In Spain	Not in Spain
Saucer on ground	53	705
Saucer hovering	38	412
Total	$n = 91$	$m = 1117$

Step 1. Hypotheses

$$H_0:\ p_S = p_{NS} \qquad \text{vs.} \qquad H_1:\ p_S \neq p_{NS}, \qquad \alpha = 0.01.$$

Step 2. Test statistic (Theorem 9.4.1)

With $x = 53$ successes in $n = 91$ and $y = 705$ in $m = 1117$, and pooled estimate $p_e = \dfrac{x+y}{n+m}$,

$$Z = \frac{\dfrac{x}{n} - \dfrac{y}{m}}{\sqrt{\dfrac{p_e(1-p_e)}{n} + \dfrac{p_e(1-p_e)}{m}}} \overset{\text{approx}}{\sim} N(0,1).$$

Step 3. Sample proportions

$$\hat{p}_S = \frac{53}{91} = 0.5824, \qquad \hat{p}_{NS} = \frac{705}{1117} = 0.6312, \qquad p_e = \frac{53+705}{91+1117} = \frac{758}{1208} = 0.6275.$$

Step 4. Standard error

$$SE = \sqrt{(0.6275)(0.3725)\left(\frac{1}{91} + \frac{1}{1117}\right)} = \sqrt{(0.23375)(0.011884)} = \sqrt{0.002778} = 0.05271.$$

Step 5. Observed Z

$$z = \frac{0.5824 - 0.6312}{0.05271} = \frac{-0.0487}{0.05271} = -0.92.$$

Step 6. Critical region and decision

Reject H_0 if $|z| \geq z_{.005} = 2.576$. Since $|-0.92| < 2.576$ (two-sided $P = 2\,P(Z \leq -0.92) \approx 0.355$), we do not reject.

Conclusion. Fail to reject H_0. There is no significant evidence at $\alpha = 0.01$ that the proportion of "on the ground" sightings differs between Spain and elsewhere.

Problem 9.5.2 (Fiddler crabs: 95% confidence interval for $\mu_X - \mu_Y$)

Given. X = % time waving, males with domes ($n = 5$); Y = males without domes ($m = 7$); $s_p = 11.2$.

Step 1. Interval formula (Theorem 9.5.1)

$$(\bar{x} - \bar{y}) \pm t_{\alpha/2,\, n+m-2}\, s_p \sqrt{\frac{1}{n} + \frac{1}{m}}.$$

Step 2. Sample means

$$\sum x_i = 100.0 + 58.6 + 93.5 + 83.6 + 84.1 = 419.8 \Rightarrow \bar{x} = 83.960,$$

$$\sum y_i = 76.4 + 84.2 + 96.5 + 88.8 + 85.3 + 79.1 + 83.6 = 593.9 \Rightarrow \bar{y} = 84.843, \qquad \bar{x} - \bar{y} = -0.883.$$

Step 3. Critical value

$1 - \alpha = 0.95$, $df = 5 + 7 - 2 = 10$, so $t_{.025,10} = 2.228$.

Step 4. Margin of error

$$\sqrt{\frac{1}{5} + \frac{1}{7}} = \sqrt{0.34286} = 0.58554, \qquad \text{ME} = 2.228\,(11.2)(0.58554) = 14.61.$$

Step 5. The interval

$$-0.883 \pm 14.61 \;=\; (-15.50,\; 13.73).$$

Conclusion. The 95% CI for $\mu_X - \mu_Y$ is $(\mathbf{-15.50}, \mathbf{13.73})$. Because it contains 0, the data do not show that a dome affects the time spent waving; this agrees with failing to reject $H_0 : \mu_X = \mu_Y$ at $\alpha = 0.05$.

Problem 9.5.6 (Twain vs. Snodgrass: 95% CI for σ_X^2/σ_Y^2)

Given. Case Study 9.2.1: X = proportion of three-letter words, Twain ($n = 8$); Y = Quintus Curtius Snodgrass ($m = 10$).

Step 1. Interval formula (Theorem 9.5.2)

$$\left(\frac{s_X^2}{s_Y^2}\, F_{\alpha/2,\, m-1,\, n-1},\;\; \frac{s_X^2}{s_Y^2}\, F_{1-\alpha/2,\, m-1,\, n-1} \right) \quad \text{for} \quad \frac{\sigma_X^2}{\sigma_Y^2}.$$

Step 2. Sample variances

$$\sum x_i = 1.855,\ \sum x_i^2 = 0.431613,\quad SS_X = 0.431613 - \frac{(1.855)^2}{8} = 0.001485,\quad s_X^2 = \frac{0.001485}{7} = 2.121 \times 10^{-4},$$
$$\sum y_i = 2.097,\ \sum y_i^2 = 0.440581,\quad SS_Y = 0.440581 - \frac{(2.097)^2}{10} = 0.000840,\quad s_Y^2 = \frac{0.000840}{9} = 9.334 \times 10^{-5}.$$
$$\frac{s_X^2}{s_Y^2} = \frac{2.121 \times 10^{-4}}{9.334 \times 10^{-5}} = 2.2725.$$

Step 3. F quantiles

Degrees of freedom $(m-1, n-1) = (9, 7)$:

$$F_{.025,9,7} = 0.2383, \qquad F_{.975,9,7} = 4.823.$$

Step 4. Endpoints

$$\text{lower} = 2.2725\,(0.2383) = 0.541, \qquad \text{upper} = 2.2725\,(4.823) = 10.96.$$

Conclusion. The 95% CI for σ_X^2/σ_Y^2 is (**0.541**, **10.96**). It contains 1, so equal variances are plausible. That agrees with the equal-variance assumption made in the t test of Case Study 9.2.1.

Problem 9.5.10 (Nightmares: 80% CI for $p_M - p_W$)

Given. Case Study 9.4.2: $x = 55$ of $n = 160$ men and $y = 60$ of $m = 192$ women reported frequent nightmares.

Step 1. Interval formula (Theorem 9.5.3)

$$\left(\frac{x}{n} - \frac{y}{m}\right) \pm z_{\alpha/2}\sqrt{\frac{\frac{x}{n}\left(1-\frac{x}{n}\right)}{n} + \frac{\frac{y}{m}\left(1-\frac{y}{m}\right)}{m}}.$$

Step 2. Point estimates

$$\hat{p}_M = \frac{55}{160} = 0.34375, \qquad \hat{p}_W = \frac{60}{192} = 0.3125, \qquad \hat{p}_M - \hat{p}_W = 0.03125.$$

Step 3. Critical value

$1 - \alpha = 0.80 \Rightarrow \alpha/2 = 0.10$, so $z_{.10} = 1.2816$.

Step 4. Standard error

$$\frac{(0.34375)(0.65625)}{160} = 0.0014099, \qquad \frac{(0.3125)(0.6875)}{192} = 0.0011190,$$

$$SE = \sqrt{0.0014099 + 0.0011190} = \sqrt{0.0025289} = 0.05029.$$

Step 5. Margin of error and interval

$$\text{ME} = 1.2816\,(0.05029) = 0.0645, \qquad 0.03125 \pm 0.0645 = (-0.033,\ 0.096).$$

Conclusion. The 80% CI for $p_M - p_W$ is $(-\mathbf{0.033}, \mathbf{0.096})$. It contains 0, consistent with the non-significant result of Case Study 9.4.2.

Chapter 10: Goodness-of-Fit Tests

Problem 10.2.6 (Baseball: multinomial probability)

Given. Per at-bat: Out .713, Single .270, Double .010, Triple .002, Home run .005. Five at-bats; find $P(2 \text{ outs, } 2 \text{ singles, } 1 \text{ double})$.

Step 1. Multinomial pdf

For n independent trials with t outcomes,

$$P(X_1 = k_1, \ldots, X_t = k_t) = \frac{n!}{k_1!\, k_2! \cdots k_t!}\, p_1^{k_1} p_2^{k_2} \cdots p_t^{k_t}.$$

Here $n = 5$, $(k_1, k_2, k_3, k_4, k_5) = (2, 2, 1, 0, 0)$, and since $0! = 1$ and $p^0 = 1$, the triple and home-run factors drop out.

Step 2. Multinomial coefficient

$$\frac{5!}{2!\, 2!\, 1!} = \frac{120}{(2)(2)(1)} = 30.$$

Step 3. Probability factors

$$(.713)^2 = 0.508369, \qquad (.270)^2 = 0.0729, \qquad (.010)^1 = 0.010.$$

Step 4. Multiply

$$P = 30\,(0.508369)(0.0729)(0.010) = 30\,(0.00037060) = 0.01112.$$

Conclusion. $P(2 \text{ outs, } 2 \text{ singles, } 1 \text{ double}) \approx \mathbf{0.0111}$.

Problem 10.3.6 (Schizophrenia and birth season: χ^2 goodness-of-fit)

Given. $n = 5139$ persons; 1383 born in the first quarter. The census-based expected number in the first quarter is 1292.1.

Step 1. Hypotheses

Let $p =$ probability that a schizophrenic is born in the first quarter, and $p_0 = 1292.1/5139 = 0.2514$.

$$H_0 : \; p = p_0 \qquad \text{vs.} \qquad H_1 : \; p \neq p_0, \qquad \alpha = 0.05.$$

Step 2. Test statistic (Theorem 10.3.1)

$$D = \sum_{i=1}^{t} \frac{(X_i - np_{i0})^2}{np_{i0}} \overset{\text{approx}}{\sim} \chi^2_{t-1},$$

valid when all expected counts are at least 5.

Step 3. Observed and expected counts ($t = 2$ cells)

	First quarter	Other quarters
Observed k_i	1383	$5139 - 1383 = 3756$
Expected np_{i0}	1292.1	$5139 - 1292.1 = 3846.9$

Step 4. Compute d

$$d = \frac{(1383 - 1292.1)^2}{1292.1} + \frac{(3756 - 3846.9)^2}{3846.9} = \frac{(90.9)^2}{1292.1} + \frac{(-90.9)^2}{3846.9} = \frac{8262.81}{1292.1} + \frac{8262.81}{3846.9} = 6.395 + 2.148 = 8.54.$$

Step 5. Critical value and decision

$df = t - 1 = 1$, and $\chi^2_{.95,1} = 3.841$. Reject H_0 if $d \geq 3.841$. Since $8.54 > 3.841$ ($P \approx 0.0035$), we reject.

Conclusion. Reject H_0 at $\alpha = 0.05$. There is significant evidence that the proportion of schizophrenics born in the first quarter differs from the general population (it is higher: $1383 > 1292.1$).

Problem 10.3.9 (Racetrack: uniform goodness-of-fit)

Given. 144 races, winners by starting post $1, \ldots, 8$: $32, 21, 19, 20, 16, 11, 14, 11$; $\alpha = 0.05$.

Step 1. Hypotheses

$$H_0: \; p_1 = p_2 = \cdots = p_8 = \tfrac{1}{8} \qquad \text{vs.} \qquad H_1: \; \text{not all } p_i \text{ equal}, \qquad \alpha = 0.05.$$

Step 2. Expected counts

$$np_{i0} = 144 \cdot \frac{1}{8} = 18 \quad \text{for every post.}$$

Step 3. Compute d

Post	1	2	3	4	5	6	7	8	Total
k_i	32	21	19	20	16	11	14	11	144
$k_i - 18$	14	3	1	2	-2	-7	-4	-7	0
$(k_i - 18)^2$	196	9	1	4	4	49	16	49	328

$$d = \sum \frac{(k_i - 18)^2}{18} = \frac{328}{18} = 18.22.$$

Step 4. Critical value and decision

$df = 8 - 1 = 7$, $\chi^2_{.95,7} = 14.067$. Since $18.22 > 14.067$ ($P \approx 0.011$), reject H_0.

Conclusion. Reject H_0 at $\alpha = 0.05$. Horses do not win equally often from every starting position; post 1 produced noticeably more winners than expected (32 vs. 18).

Problem 10.4.10 (Turnovers: Poisson goodness-of-fit, λ unknown)

Given. $N = 440$ games; number of games with k turnovers: $75, 125, 126, 60, 34, 13, 7$ for $k = 0, 1, 2, 3, 4, 5, 6^+$; $\alpha = 0.05$.

Step 1. Hypotheses

$$H_0: \; p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \; k = 0, 1, 2, \ldots \qquad \text{vs.} \qquad H_1: \text{ data are not Poisson.}$$

Step 2. Estimate λ (maximum likelihood = sample mean)

Treating "6^+" as 6,

$$\sum k\, n_k = 0(75)+1(125)+2(126)+3(60)+4(34)+5(13)+6(7) = 125+252+180+136+65+42 = 800,$$

$$\hat{\lambda} = \frac{800}{440} = 1.8182.$$

Step 3. Estimated cell probabilities

$$\hat{p}_0 = e^{-1.8182} = 0.16232, \qquad \hat{p}_k = \hat{p}_{k-1} \cdot \frac{\hat{\lambda}}{k},$$

$$\hat{p}_1 = 0.29513, \; \hat{p}_2 = 0.26830, \; \hat{p}_3 = 0.16261, \; \hat{p}_4 = 0.07391, \; \hat{p}_5 = 0.02688,$$

$$\hat{p}_{6^+} = 1 - \sum_{k=0}^{5} \hat{p}_k = 1 - 0.98915 = 0.01085.$$

Step 4. Expected counts $440\,\hat{p}_k$ and pooling

The expected count in the 6^+ cell is $4.78 < 5$, so we pool the cells for $k = 5$ and $k = 6^+$ (observed $13 + 7 = 20$, expected $11.83 + 4.78 = 16.60$).

Turnovers	0	1	2	3	4	5^+
Observed k_i	75	125	126	60	34	20
Expected $440\hat{p}_i$	71.42	129.86	118.05	71.55	32.52	16.60
$\frac{(k_i - 440\hat{p}_i)^2}{440\hat{p}_i}$	0.179	0.182	0.535	1.864	0.067	0.695

Step 5. Test statistic

$$d_1 = 0.179 + 0.182 + 0.535 + 1.864 + 0.067 + 0.695 = 3.52.$$

Step 6. Degrees of freedom and decision

Number of cells $t = 6$; one parameter estimated ($s = 1$), so

$$df = t - 1 - s = 6 - 1 - 1 = 4, \qquad \chi^2_{.95,4} = 9.488.$$

Reject H_0 if $d_1 \geq 9.488$. Since $3.52 < 9.488$ ($P \approx 0.47$), do not reject.

Conclusion. Fail to reject H_0 at $\alpha = 0.05$. A Poisson model is consistent with the turnover data. (Without pooling: $d_1 = 3.98$ on 5 df, critical value 11.07, the same conclusion.)

Problem 10.5.6 (Blood pressure: χ^2 test of independence)

Given. 92 father–child pairs classified by blood pressure third; $\alpha = 0.05$.

	Child lower	Child middle	Child upper	Row total
Father lower	14	11	8	33
Father middle	11	11	9	31
Father upper	6	10	12	28
Column total	31	32	29	92

Step 1. Hypotheses

H_0 : child's and father's blood pressure categories are independent vs. H_1 : they are not independent.

Step 2. Test statistic (Theorem 10.5.1)

Under H_0, the expected count in cell (i, j) is $\hat{e}_{ij} = \dfrac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{N}$, and

$$D = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(X_{ij} - \hat{e}_{ij})^2}{\hat{e}_{ij}} \overset{\text{approx}}{\sim} \chi^2_{(r-1)(c-1)}.$$

Step 3. Expected counts

$$\hat{e}_{11} = \frac{33 \cdot 31}{92} = 11.12, \quad \hat{e}_{12} = \frac{33 \cdot 32}{92} = 11.48, \quad \hat{e}_{13} = \frac{33 \cdot 29}{92} = 10.40,$$
$$\hat{e}_{21} = \frac{31 \cdot 31}{92} = 10.45, \quad \hat{e}_{22} = \frac{31 \cdot 32}{92} = 10.78, \quad \hat{e}_{23} = \frac{31 \cdot 29}{92} = 9.77,$$
$$\hat{e}_{31} = \frac{28 \cdot 31}{92} = 9.43, \quad \hat{e}_{32} = \frac{28 \cdot 32}{92} = 9.74, \quad \hat{e}_{33} = \frac{28 \cdot 29}{92} = 8.83.$$

All expected counts exceed 5.

Step 4. Cell contributions $\frac{(O-E)^2}{E}$

	Lower	Middle	Upper
Father lower	$\frac{(14-11.12)^2}{11.12} = 0.746$	$\frac{(11-11.48)^2}{11.48} = 0.020$	$\frac{(8-10.40)^2}{10.40} = 0.555$
Father middle	$\frac{(11-10.45)^2}{10.45} = 0.029$	$\frac{(11-10.78)^2}{10.78} = 0.004$	$\frac{(9-9.77)^2}{9.77} = 0.061$
Father upper	$\frac{(6-9.43)^2}{9.43} = 1.250$	$\frac{(10-9.74)^2}{9.74} = 0.007$	$\frac{(12-8.83)^2}{8.83} = 1.141$

Step 5. Sum and compare

$$d = 0.746 + 0.020 + 0.555 + 0.029 + 0.004 + 0.061 + 1.250 + 0.007 + 1.141 = 3.81.$$

$df = (3-1)(3-1) = 4$, and $\chi^2_{.95,4} = 9.488$.

Step 6. Decision

Since $3.81 < 9.488$ ($P \approx 0.43$), we do not reject H_0.

Conclusion. Fail to reject H_0 at $\alpha = 0.05$. There is no significant evidence that a child's blood pressure category is related to the father's, so a father's category is not shown to be useful for screening children.